THE APPROXIMATION PROPERTY REDUCES TO ALGEBRAS GEOMETRICALLY REGULAR OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-product is same-chapter NOT carded (dropped in judgment: Omitted with Hint) and is declared external; input lemma-lifting is published (tag 07CP).

Allowed standard prerequisites (editorial).

- Regular maps are stable under quotient base change. More on Algebra, lemma regular-base-change (cross-chapter, named). Keeps R/I -> Lambda/ILambda regular along the Noetherian induction.
- Ind-smoothness is stable under infinitesimal deformations. proposition-lift (carded L26807). Shows the maximal counterexample ring R is reduced.
- Products of PT situations. lemma-product (same chapter, NOT carded, declared external). Transfers PT from the fields K_1 x ... x K_n to S^{-1}R -> S^{-1}Lambda.
- Spreading a smooth factorization out of a localization. lemma-delocalize-base (carded L26810). Produces the factorization with an elementary standard denominator pi in S.
- Lifting along annihilator-controlled elements. lemma-lifting (same chapter, PUBLISHED 07CP). Lifts the mod-pi^8 factorization to a smooth D over R with a mod-pi^4 factorization.
- Desingularization with compatible mod-pi^4 maps. lemma-desingularize-strictly-standard (carded L26808). Final step: converts the mod-pi^4 compatibility into an honest smooth factorization.

Source reference. Original reference 00FR: focused-seeds/geometry/S9-smo-reduce-to-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02LX: focused-seeds/geometry/S9-smo-reduce-to-field/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07C3: focused-seeds/geometry/S9-smo-reduce-to-field/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07F4: sources/stacks-smoothing/pinned-original-source.tex, source lines 2154–2154. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CT: sources/stacks-smoothing/pinned-original-source.tex, source lines 2029–2029. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CP: sources/stacks-smoothing/pinned-original-source.tex, source lines 1638–1638. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07F3: sources/stacks-smoothing/pinned-original-source.tex, source lines 2143–2143. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07C1: focused-seeds/geometry/S9-smo-reduce-to-field/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07CM: sources/stacks-smoothing/pinned-original-source.tex, source lines 1591–1591. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07F2: sources/stacks-smoothing/pinned-original-source.tex, source lines 2133–2133. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *If for every Situation 07F2 where R is a field PT holds, then PT holds in general.*

Proof. Assume PT holds for any Situation 07F2 where R is a field. Let $R \to \Lambda$ be as in Situation 07F2 arbitrary. Note that $R/I \to \Lambda/I\Lambda$ is another regular ring map of Noetherian rings, see More on Algebra, Lemma 07C1. Consider the set of ideals

$$\mathcal{I} = \{I \subset R \mid R/I \to \Lambda/I\Lambda \text{ does not have PT}\}$$

We have to show that $\mathcal{I}$ is empty. If this set is nonempty, then it contains a maximal element because R is Noetherian. Replacing R by R/I and Λ by Λ/I we obtain a

situation where PT holds for $R/I \to \Lambda/I\Lambda$ for any nonzero ideal of R. In particular, we see by applying Proposition 07CM that R is a reduced ring.

Let $A \to \Lambda$ be an R-algebra homomorphism with A of finite presentation. We have to find a factorization $A \to B \to \Lambda$ with B smooth over R, see Algebra, Lemma 07C3.

Let $S \subset R$ be the set of nonzerodivisors and consider the total ring of fractions $Q = S^{-1}R$ of R. We know that $Q = K_1 \times \ldots \times K_n$ is a product of fields, see Algebra, Lemmas 02LX and 00FR. By Lemma 07F3 and our assumption PT holds for the ring map $S^{-1}R \to S^{-1}\Lambda$. Hence we can find a factorization $S^{-1}A \to B' \to S^{-1}\Lambda$ with B' smooth over $S^{-1}R$.

We apply Lemma 07F4 and find a factorization $A \to B \to \Lambda$ such that some $\pi \in S$ is elementary standard in B over R. After replacing A by B we may assume that π is elementary standard, hence strictly standard in A. We know that $R/\pi^8R \to \Lambda/\pi^8\Lambda$ satisfies PT. Hence we can find a factorization $R/\pi^8R \to A/\pi^8A \to \bar{C} \to \Lambda/\pi^8\Lambda$ with $R/\pi^8R \to \bar{C}$ smooth. By Lemma 07CP we can find an R-algebra map $D \to \Lambda$ with D smooth over R and a factorization $R/\pi^4R \to A/\pi^4A \to D/\pi^4D \to \Lambda/\pi^4\Lambda$. By Lemma 07CT we can find $A \to B \to \Lambda$ with B smooth over R which finishes the proof. $\square$